

A cross-dataset based zero-day intrusion detection system by integrating siamese network and reinforcement learning

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ABSTRACT

Zero-day attacks threaten IoT security as signature-based detection fails against novel exploits. This paper proposes a hybrid Intrusion Detection System integrating unsupervised anomaly detection, non-parametric Siamese-based cross-dataset dissimilarity filtering, and Proximal Policy Optimization (PPO)-based adaptive defense. Unsupervised models isolate anomalous traffic, Siamese-based correlation extracts structurally rare zero-day candidates, and the PPO agent learns optimal defense policies via environmental feedback. Evaluations on CIC-IoT-2023 and CIC-BCCC-NRC-TabularIoTAttacks-2024 demonstrate 99.28% training accuracy, 99.07% unseen attack accuracy, and 93.94% zero-day detection rate with 0.50 ms latency and 2.21% false-positive rate, providing a scalable, proactive, self-learning defense architecture for autonomous IoT cybersecurity.

1. Introduction

The rapid expansion and increasing connectivity of IoT networks have heightened vulnerabilities for individuals, businesses, and governments to cyberattacks. Among these, zero-day attacks are particularly dangerous, exploiting unknown software or hardware vulnerabilities before patches are available. The established intrusion detecting mechanisms (IDS) that are mostly based on prewritten attack signature are not able to handle such dynamic threats as attackers often alter the attack signature to avoid detection [1,2]. As a remedy to the limitations of signature-based methods, anomaly-based detection methods have been proposed, which are founded on differences between normal system behavior and the attacks. Although these methods provide a potential solution in the detection of unknown attacks, they tend to be typified by a high false-positive rate especially in dynamic IoT systems. False alarms are many and they overwhelm the security analyst reducing the reliability of the system in the instances of real-life deployment [3]. Further, recent advances in artificial intelligence, specifically RL, present a potential solution to adaptive cybersecurity systems (even though these methods are difficult to detect at zero-days) [4,5]. Through constant interaction with the environment, directed by the environment, RL techniques are capable of processing large amounts of network data and train to employ optimal defense strategies learned by the environment. The algorithms, PPO and Deep

Q-Networks (DQN), allow the systems to react to unknown attack patterns and address a fundamental drawback of a pattern-based attack detector [6]. The overall goal of this research is to come up with a hybrid zero-day intrusion detection model for IoT networks that integrates unsupervised anomaly detection, cross-dataset anomaly correlation using a non-parametric Siamese networks, and reinforcement learning-based adaptive defense mechanisms. The key results of the research are to show high performance with the use of unlabeled data, anomaly-based detection, and RL based on adaptive defensive policy in the work of the IoT based on the zero-day attacks, operation on a large scale, and in real-time, overcoming the main drawbacks of the current cybersecurity-focused applications; to create a three-layer framework of the discussed system with intent to integrate unsupervised anomaly detection, Siamese similarity analysis, and adaptive RL response to such data; to create a non-parametric Siamese network using Euclidean distance.

2. Related works

Existing intrusion detection systems for IoT environments can be broadly categorized into deep learning-based, ensemble-based, and zero-shot or unsupervised approaches, each addressing different aspects of detection performance, scalability, and generalization. Wahed

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et al. [7] introduced an AI-enhanced IDS using RNN, LSTM, and GRU for sequential traffic analysis. The architecture processes real-time data with approximately 2-3 million records per attack type, achieving 98.5% accuracy and maintaining false positive rates below 8%. It integrates NLP with optimized Random Forests for automated threat response but is computationally intensive and requires retraining in resource-limited environments. Zaki et al. [8] developed an ensemble IDS for IoT zero-day attack detection using sparse random projection with KNN, XGBoost, and SGD classifiers. Evaluated on CIC-DDoS2019 and CICIDS2017 datasets, the system achieved 99.91% accuracy with 0.22-second detection time through GridSearchCV hyperparameter optimization. Despite the high precision in attack detection by these approaches, they generally rely on large-scale labeled datasets and incur significant computational overhead, limiting their applicability in dynamic and resource-constrained IoT environments. Sarhan et al. [9] introduced a zero-shot learning framework addressing signature-based NIDS limitations through semantic attribute learning. The methodology trains ML models on known attacks (phishing, DDoS, XSS, SQL injection) to generalize zero-day attack detection. Using MLP and RF models, the system achieved 95% detection rate for generic attacks but struggled with complex reconnaissance and exploit patterns (90% ZDR). Roopak et al. [10] proposed an unsupervised IDS for IoT zero-day DDoS detection using Gaussian Random Projection for dimensionality reduction (87 to 25 dimensions). The ensemble method combines K-means, GMM, and SGD-OCSVM through hard voting, achieving 94.5% accuracy and 94.3% F1-score on CIC-DDoS2019. The ensemble approach is more computationally expensive and requires large tuning although it is capable of handling high-dimensional unlabeled data. Abbas-Escribano and Debar [11] implemented an enhanced high-interaction honeypot system based on MITRE ATTACK taxonomy for attack analysis. The virtualized multilevel architecture recorded 1.5 million events over 17 days, successfully logging attacker behavior and generating IoCs while preventing deeper system penetration. Nonetheless, the exposure period was brief and the threat could not be analyzed extensively due to the limited capacity to attract advanced attackers. Although these approaches improve zero-day detection capability, they remain sensitive to feature representation, class imbalance, and distributional shifts, which reduces robustness across heterogeneous datasets. While these methods provide strong detection performance, they primarily rely on static decision mechanisms and lack adaptability in evolving threat environments. In this regard, recent research based on reinforcement learning by Zhao et al. [5,12] demonstrates that policy gradient methods such as PPO enable stable and adaptive decision-making through continuous interaction with dynamic environments. However, most RL-based IDS frameworks are designed for domain-specific scenarios, such as vehicular or UAV networks, limiting their applicability to heterogeneous IoT systems. Additionally, similarity-based approaches inspired by Siamese neural networks have been explored to capture structural relationships between data samples. However, deep Siamese models require computationally intensive training and are sensitive to data imbalance. This reveals a research gap in developing a lightweight and adaptive framework that combines cross-dataset anomaly correlation with reinforcement learning for generalized IoT security. The proposed study addresses this gap by integrating PPO-based adaptive learning with non-parametric Siamese-based similarity analysis.

3. Methodology

An unsupervised multi-stage IDS framework is proposed for large-scale IoT networks, detecting known and zero-day threats via high-confidence anomaly filtering and RL-based adaptive defense. As shown in Fig. 1, the pipeline integrates dataset-specific unsupervised anomaly detection, cross-dataset Siamese distance-based validation, and RL response optimization, progressively refining traffic to isolate structurally rare, semantically dissimilar attack behaviors for decision learning without signature-based methods.

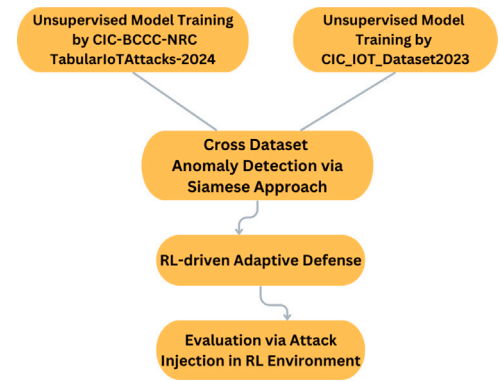


Fig. 1. Workflow of the Hybrid Zero-Day IDS

Table 1

Evaluation metrics of unsupervised models on both datasets.

Model	CIC-IoT-2023			CIC-BCCC-NRC-2024		
	Acc.	F1	Recall	Acc.	F1	Recall
PCA	0.88	0.87	0.88	0.98	0.98	0.98
Autoencoder	0.89	0.88	0.89	0.97	0.98	0.98
Isolation Forest	0.88	0.88	0.88	0.95	0.95	0.95
One-Class SVM	0.89	0.88	0.88	0.94	0.96	0.97
K-means	0.88	0.88	0.88	0.97	0.98	0.98

3.1. Dataset description

Experiments utilize CIC-IoT-2023 and CIC-BCCC-NRC-TabularIoTAttacks-2024 datasets taken from the Canadian Institute for Cybersecurity.

3.2. Data preprocessing

Prior to model training, raw IoT network data were preprocessed to ensure cross-dataset compatibility. Infinite values and duplicate records were removed. Class labels were retained for evaluation but excluded during training to satisfy unsupervised learning requirements. Numerical features were standardized, and categorical protocol attributes were encoded. Feature schemas of the CIC-IoT-2023 and CIC-BCCC-NRC-TabularIoTAttacks-2024 datasets were semantically aligned, yielding approximately twenty shared features.

3.3. Unsupervised anomaly detection by dataset

Algorithm 1 layer detects anomalous traffic patterns individually on each dataset. Five unsupervised algorithms were evaluated: PCA-based anomaly detection, standard autoencoders, Isolation Forest, One-Class SVM, and K-Means. All models were trained solely on benign traffic to model normal IoT behavior, flagging deviations as anomalies. Autoencoder was selected for CIC-IoT-2023 (superior reconstruction accuracy) and PCA for CIC-BCCC-NRC-2024 (high-dimensional robustness) (Table 1).

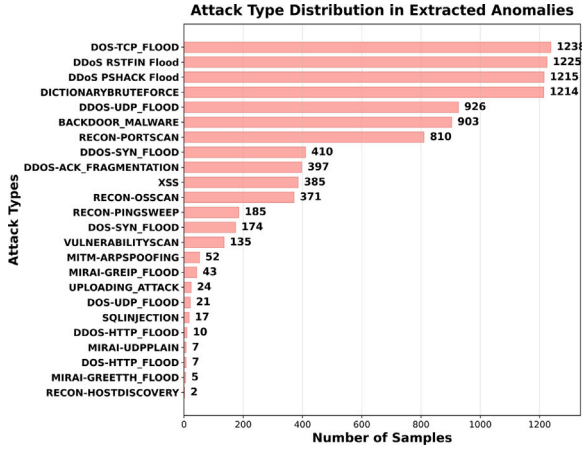
3.4. Cross-dataset anomaly filtering via Siamese distance analysis

Anomalies extracted from both datasets were compared using a non-parametric Siamese architecture with distance metrics, namely Euclidean, Manhattan, and Cosine, to identify structurally novel attack behaviors. The distance metrics were evaluated using ROC-AUC, F1-score, and distribution spread, as summarized in Table 2. Among the evaluated metrics, Euclidean distance demonstrated the highest relative separability between shared and structurally distinct anomalies. This

Table 2

Evaluation metrics of the three distance-based similarity models.

Metric	AUC	F1-Score	Spread
Euclidean	0.5048	0.3323	170026.6319
Manhattan	0.5046	0.3323	170022.4141
Cosine	0.4985	0.3320	0.0778

**Fig. 2.** 24 types of attacks in extracted anomalies.

stage acts as a relative anomaly filtering tool instead of a separate classifier, focusing on identifying uncommon patterns within overlapping distributions. Although near-random AUC values were observed due to class imbalance and cross-dataset distributional overlap, Euclidean distance consistently outperformed the alternatives and was therefore selected for final anomaly filtering.

A distance-based threshold was subsequently applied to extract high-fidelity anomalies, isolating structurally distinct attack patterns, as illustrated in Fig. 2. The resulting set of 24 attack types constitutes the exclusive input for downstream reinforcement learning. The complete cross-dataset anomaly correlation and filtering procedure is detailed in Algorithm 2. This stage follows a dual-input comparative structure based on the Siamese paradigm, where paired inputs from two datasets are compared under a common distance function without a parameterized neural embedding model.

3.5. Reinforcement learning-driven adaptive defense

Algorithm 3 employs reinforcement learning to derive adaptive response strategies from the filtered anomaly set. Three RL agents: PPO, DQN, and SAC were evaluated, with state and action spaces defined by anomaly characteristics and defensive response models. The reward function is defined as:

$$R_t = \begin{cases} +1 & \text{if the detection is correct} \\ -1 & \text{if the detection is incorrect} \end{cases} \quad (1)$$

This sparse binary reward directly rewards the correct intrusion detection and penalizes both false positives and missed attacks equally. Based on superior stability and generalization performance, PPO was selected as the optimal agent. The learned policy was evaluated using both seen and unseen anomalies. Together, Algorithms 1, 2, and 3 constitute a fully unsupervised, scalable framework for proactive zero-day threat mitigation without reliance on prior attack signatures.

Algorithm 1 Unsupervised Anomaly Model Training

Input: Training data D_{train} , testing data D_{test}

Output: Extracted anomaly set A

- 1: Remove labels from D_{train} and clean missing or infinite values
- 2: Normalize all features using standard scaling
- 3: Train Autoencoder f_θ on D_{train} by minimizing reconstruction loss
- 4: Compute reconstruction error for each test sample:

$$e_i = \|x_i - f_\theta(x_i)\|^2$$

- 5: Apply PCA on D_{train} and project D_{test} onto principal subspace
- 6: Compute PCA residual error for each test sample
- 7: Fuse anomaly scores from Autoencoder and PCA
- 8: Determine anomaly threshold τ using F1-score maximization
- 9: Mark x_i as anomalous if anomaly score(x_i) $> \tau$
- 10: Aggregate all detected anomalies into set A

Algorithm 2 Cross-Dataset Anomaly Correlation Using Non-Parametric Siamese Architecture

Input: Anomaly sets A_1, A_2

Output: Zero-day anomaly set Z

- 1: Initialize distance functions $D \in \{\text{Euclidean, Manhattan, Cosine}\}$
- 2: **for** each $a_i \in A_1$ **do**
- 3: **for** each $a_j \in A_2$ **do**
- 4: Compute distance: $d_{ij} = D(a_i, a_j)$
- 5: Mark (a_i, a_j) as dissimilar if $d_{ij} > \delta$
- 6: **end for**
- 7: **end for**
- 8: Aggregate structurally dissimilar anomalies
- 9: Identify aggregated anomalies as zero-day candidates Z

4. Results and evaluation

A dual-dataset evaluation strategy assesses zero-day detection performance where a combined evaluation metric in which training performance is weighted by 40 percent and unseen data is weighted by 60 percent to emphasize the use of generalization to detect zero-day attacks and guide hyperparameter selection to achieve stable performance on known and unseen attack distributions. PPO outperforms DQN/SAC across all metrics

Algorithm 3 RL-Based Adaptive Defense

Input: Zero-day anomaly set Z , environment $\mathcal{E}$

Output: Optimal defense policy π^*

- 1: Define state space S from anomaly features
- 2: Define action space A as mitigation responses
- 3: Initialize policy network π_θ and value network V_ϕ
- 4: **for** each training episode **do**
- 5: Observe current state s_t
- 6: Select action $a_t \sim \pi_\theta(s_t)$
- 7: Execute a_t in $\mathcal{E}$ and observe reward r_t and s_{t+1}
- 8: Store transition (s_t, a_t, r_t, s_{t+1})
- 9: **end for**
- 10: Update policy parameters using PPO optimization
- 11: Evaluate policy performance using DQN and SAC baselines
- 12: Derive optimal policy: $\pi^*(s) = \arg \max_a Q^*(s, a)$

Table 3. Following ablation, PPO was optimized with learning rate 1×10^{-3} , $\gamma = 0.99$, and $\epsilon = 0.15$ over 300 episodes, achieving optimal performance.

4.1. Performance evaluation and zero-day performance

Evaluation was conducted using the best-performing reinforcement learning configuration, PPO. A simple reward scheme assigns positive rewards for correct detections and penalties for misclassifications, while zero-day samples are identified using an empirically optimized distance-based threshold. As summarized in Table 3, the proposed model achieves strong and consistent performance on both seen and

Table 3
PPO performance metrics and RL algorithm comparison.

Metric	PPO (Optimal)		vs DQN/SAC
	Training	Unseen	
Accuracy	99.28%	99.07%	98.8%/96.32%
Precision	99.30%	99.16%	–
F1-Score	99.29%	99.10%	–
Zero-Day Detection	93.94%	–	93.1%/84.1%
ROC-AUC	98.83%	99.69%	–
FPR	0.79%	2.21%	–
Latency	0.50 ms	–	0.65/0.601 ms

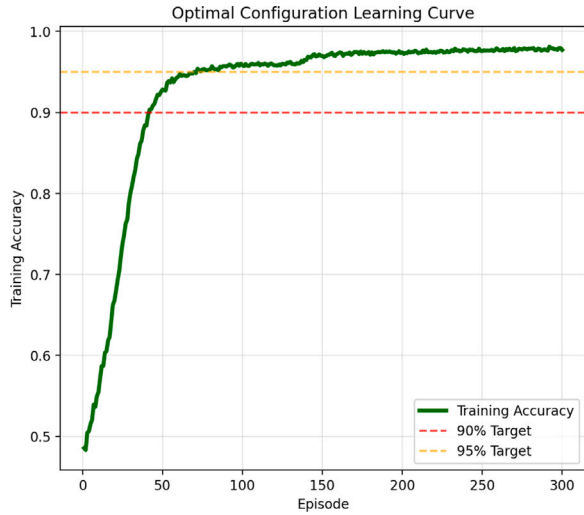


Fig. 3. PPO Agent Learning Curve.

unseen data with a minimal generalization gap and low false positive rate. Operational feasibility is confirmed through low-latency inference. As shown in Fig. 3, the PPO agent converges rapidly, reaching stable performance within a limited number of episodes. High ROC-AUC values further validate the effectiveness of the learned policy, while comparisons with DQN and SAC confirm PPO's superior stability and generalization.

4.2. Analysis of variance (ANOVA)

Factorial ANOVA revealed significant hyperparameter effects: **Learning Rate** $F(4295) = 892.3$, $p < 0.001$, $\eta^2 = 0.923$; **Clip Epsilon** $F(4295) = 245.7$, $p < 0.001$, $\eta^2 = 0.769$; **Episodes** $F(4295) = 156.2$, $p < 0.001$, $\eta^2 = 0.679$; **Gamma** $F(4295) = 89.4$, $p < 0.001$, $\eta^2 = 0.548$. Learning rate showed greatest impact, consistent with ablation rankings (Figs. 4, 5, 6).

4.3. Comparative analysis of literature

This Table 4 documents quantitative parameters for RL-based zero-day detection papers as reported by respective authors. Where a metric is not provided it is marked as “—”

5. Discussion and conclusion

The experimental outcomes demonstrate that the proposed hybrid system achieves high detection accuracy and efficiency. The PPO-based adaptive defense achieved 99.07% accuracy on unknown attacks and 93.94% zero-day detection rate, proving its capability to identify previously unseen threats. The low performance gap between training and unseen data (0.21) indicates strong generalization in dynamic IoT environments. Inference latency of 0.50 ms per sample enables

real-time deployment in high-throughput IoT and enterprise networks, while rapid convergence (90% accuracy in 42 episodes) demonstrates adaptability to novel attack patterns. The framework advances threat identification from fixed signature-based to adaptive, intelligent defense. Cross-dataset anomaly correlation via reinforcement learning enables proactive zero-day detection with a low false-positive rate of 2.21%, reducing analyst workload and enhancing confidence in automated security operations. Comparative analysis reveals PPO's superiority over DQN and SAC in policy stability and attack adaptation, while Siamese similarity analysis improves anomaly structural separation across heterogeneous data sources. More recent RL-based IDS frameworks such as TROVE by Guo et al. [19] and CCID-CAN by Sun et al. [20] demonstrate adaptive security in vehicular networks using trust-based reinforcement learning and cross-chain deep learning mechanisms, respectively. However, these approaches remain domain-specific. In contrast, the proposed framework targets heterogeneous IoT environments, enabling more generalized zero-day detection through PPO-based policy optimization and Siamese-based similarity analysis. Despite excellent results, the model depends on training data quality and representativeness, potentially limiting generalization in unfamiliar network settings. Hyperparameter tuning was performed via systematic grid search with weighted scoring, as full automation remains a limitation. Robustness against adversarial evasion attacks remains untested. Future work includes exploring adversarial evasion attacks and extending the framework to multi-agent reinforcement learning for distributed IoT defense. The modular design enables integration with federated and transfer learning, where decentralized nodes can locally train models and share updates for privacy preservation. These characteristics support scalability for large-scale IoT deployments.

5.1. Conclusion

This paper presents a hybrid zero-day IDS integrating unsupervised anomaly detection, cross-dataset Siamese similarity, and RL-based adaptive defense for IoT networks. The system has shown high cross-dataset generalization between CIC-IoT-2023 and CIC-BCCC-NRC-TabularIoTAttacks-2024 with high detection accuracy on both unseen and zero-day attacks with minimal inference latency that can be integrated into real-time operation. The findings of the experiment demonstrated the effectiveness of PPO as an adaptive defense strategy, beating other RL methods in terms of the convergence rate, detection rate, and resource use. In general, the offered three-layer architecture offers a scaled and versatile platform of autonomous cybersecurity solutions that can respond to the changing zero-day threats.

CRediT authorship contribution statement

Md. Meheraj Hossain: Writing – original draft, Methodology, Conceptualization. **Saumik Das Turja:** Writing – original draft, Visualization, Formal analysis, Data curation. **Sibgatullah Tasnim:** Writing – original draft, Visualization, Validation, Methodology. **Md. Fahmid-Ul-Alam Juboraj:** Writing – original draft, Visualization, Validation, Project administration. **Muhammad Iqbal Hossain:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

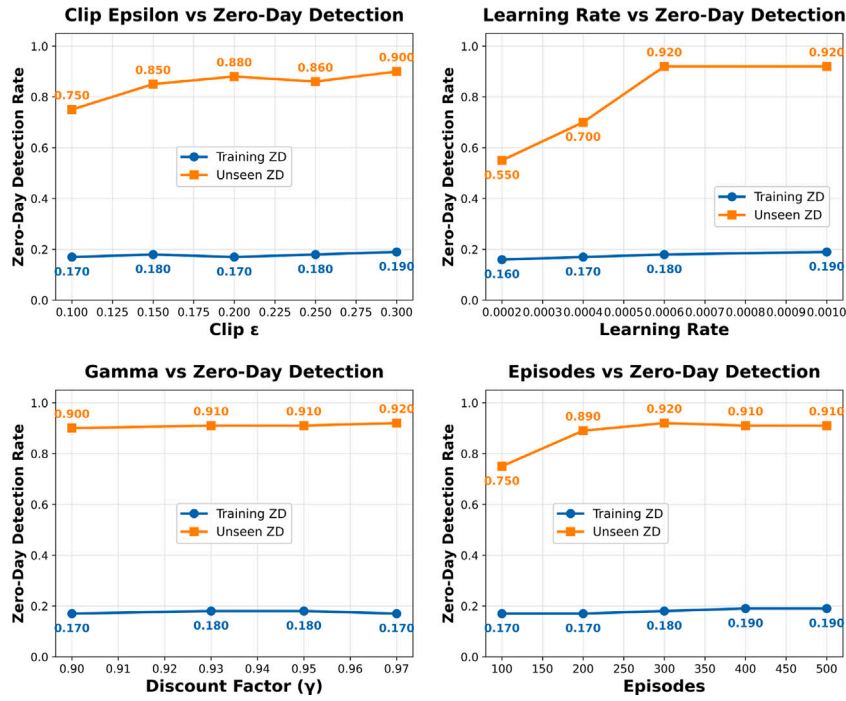


Fig. 4. Zero Day detection comparison of Hyperparameters.

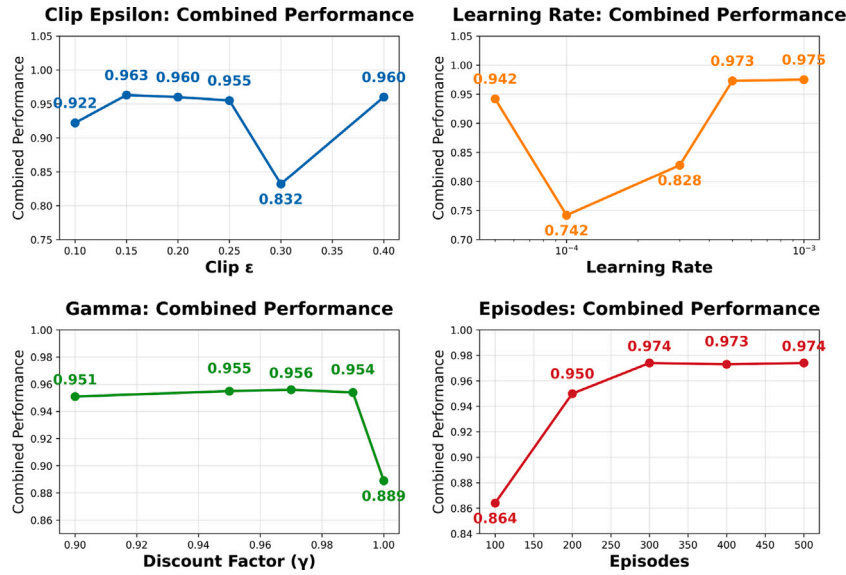


Fig. 5. Combined Performance Score of Hyperparameters.

Table 4

Comparative evaluation table of recent IDS models from contemporary literature.

No.	Study	Dataset	Acc.	F1	Latency
1	Proposed	CIC-IoT 2023	99.07	99.09	0.5 ms
2	Wu et al. [13] (2024)	NSL-KDD	98.72	98.70	–
3	Hossain [14] (2023)	UNSW-NB15	93.28	94.06	–
4	Alam et al. [15] (2023)	CICIoT2023	97.18	98.52	–
5	Dos Santos et al. [3] (2022)	NF-BoT-IoT, Mul.	99	99	–
6	Ren et al. [16] (2023)	CSE-CIC-IDS 2018	96.2	94.89	32.9 ms
7	Suresh and Jose [17] (2025)	CSE-CIC-IDS	96.8	96.3	34.3 ms
8	Alzaylaee [18] (2025)	NSL-KDD	99.1	99.1	–
		SDN-Intrusion	97.16	97	–
			92.8	91.8	–

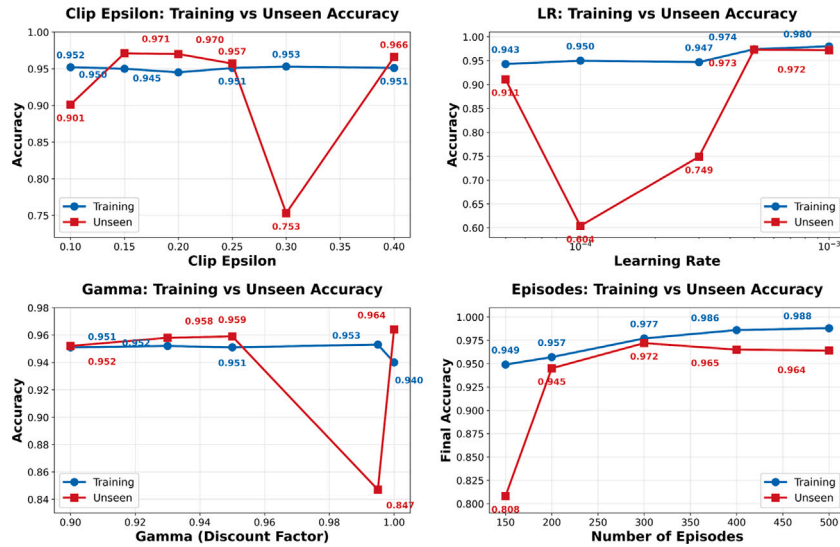


Fig. 6. Training Vs Unseen Accuracy of Hyperparameters.

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